PREDICT

WEEK 8

Capstone: Cube Maze

20 × 20 × 20 — design + solve it

최종 프로젝트: 큐브 미로를 직접 설계하고 풀기 (영어코딩)

BIG PICTURE

Everything, in one project

while + if/else + and/or + interact — all working together.

8주 동안 배운 것을 모두 합쳐요

HOW TO COUNT

Count the axis — one block, one number

1. Stand on the start block — that spot is `(0, 0, 0)`.
2. **x** → walk **right**. Add **+1** for every block.
3. **y** → jump **up**. Add **+1** for every block.
4. **z** → walk **forward**. Add **+1** for every block.

Say them in order every time: `(x, y, z)` = **across, up, forward**.

한 칸 움직이면 +1 — 오른쪽 x, 위로 y, 앞으로 z. 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft ⌨️

x	y	z
across →	up ↑	forward ⤴

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press `C` in the world)
- ✓ Type the numbers in order — `(x, y, z)`
- ✓ Press **Run** ▶ — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

Your maze must have...

20×20×20 cube the structure	2+ floors go up between them
1+ redstone door opened by a lever	start → finish a solvable path

큐브, 여러 층, 문, 레버, 시작과 끝

CONCEPT

A function packages steps

```
def go_to_next_floor():
    while not agent.detect(BLOCK, FORWARD):
        agent.move(FORWARD, 1)
    agent.move(UP, 1)
```

Name a chunk of steps once, then reuse it on every floor.

반복되는 동작을 함수로 묶어요

CODE

The solving strategy

```
while not at_goal:
    if agent.detect(BLOCK, FORWARD) or closed_door:
        agent.turn(RIGHT_TURN)
    elif lever_ahead:
        agent.interact(FORWARD)
    else:
        agent.move(FORWARD, 1)
```

Blocked → turn. Lever → flip. Open → move. Repeat until the goal.

WATCH IT!

The loop must end at the goal

```
at_goal = False
```

```
not at_goal?
```

```
turn / interact / move
```

```
▸ Agent
```

PREDICT 🤔

Why a function here?

Predict first — then click reveal

```
def go_to_next_floor():
    while not agent.detect(BLOCK, FORWARD):
        agent.move(FORWARD, 1)
    agent.move(UP, 1)
```

Reveal output 🔍 reveal

VOCAB

New words today

`function`	named, reusable group of steps	함수
`capstone`	the final project that uses everything	최종 과제
`strategy`	your plan for solving the maze	전략
`floor`	one level of the 3D cube maze	층

🐛 DEBUG

What's wrong with this code?

Goal: stop when the agent reaches the goal.

buggy

```
while True:
    if agent.detect(BLOCK, FORWARD):
        agent.turn(RIGHT_TURN)
    else:
        agent.move(FORWARD, 1)
```

Think first — then click next.

DEBUG

The bug

Bug: `while True` never stops, so the agent runs forever — even after reaching the goal. Loop on a condition that becomes false at the end instead.

fixed

```
while not at_goal:
    if agent.detect(BLOCK, FORWARD):
        agent.turn(RIGHT_TURN)
    else:
        agent.move(FORWARD, 1)
```

YOUR TURN!

Final project 🏆

- ✓ Design a 20×20×20 cube with 2+ floors
- ✓ Add at least 1 redstone door and 1 lever
- ✓ Mark a clear start and finish
- ✓ Solve it with `while` + `if/else` + `interact`
- ✓ Send your full walkthrough video on KakaoTalk

QUIZ

QUESTION 1 OF 3

Why define `go_to_next_floor()` as a function?

- To make the agent move faster.
- Functions are required by Minecraft.
- To hide the code.
- To reuse the same steps on every floor without rewriting them.

QUIZ

QUESTION 2 OF 3

What's the danger of `while True` for a maze solver?

- It only runs once.
- It runs forever and never stops at the goal.
- It turns the agent around.
- It places blocks.

QUIZ

What does a **3D** maze need that a flat one doesn't?

- A way to move up to the next floor.
- No conditions at all.
- Fewer steps.
- Only left turns.

QUIZ

Your score

0 / 3

📖 Re-read the slides. You'll get it.

↺ retake

🏆

Course complete!

You can solve any maze now. 🎉

미로 챌린지 8주 완주를 축하해요!